Random Walk Laplacian and Diffusion Map

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Introduction

Definitions

Definitions: Given a graph $G = (V, E)$, where V is the set of vertices, and E is the set of edges. Define the following matrices:

Definitions (cont.)

- The adjacency matrix, A, whose entries $A(a, b)$ are given by:

$$A(a,b) = \begin{cases} w_{a,b} & \text{if } (a,b) \in E \\ 0 & \text{otherwise} \end{cases}$$

  where $w_{a,b}$ is the weight [1] of edge (a, b). [2]
- For the scope of this topic, we will assume that A is symmetric, i.e. $w_{a,b} = w_{b,a}$ for all $a, b \in V$.

[1] One can think of the weight $w_{a,b}$ as a measure of how strongly two vertices a and b are connected.

[2] In an unweighted graph, we have $w_{a,b} = 1$ for all $(a, b) \in E$.

Definitions (cont.)

- The degree matrix, D, is the diagonal matrix in which $D(a, a)$ is the degree of vertex a.
- Algebraically, we have:

$$D = \text{diag}(A\mathbf{1})$$

where $\mathbf{1}$ is the vector with all ones.

Definitions (cont.)

- The Walk matrix, P, is then calculated by:

$$P = D^{-1}A = \begin{cases} w_{a,b}/d(a) & \text{if } (a,b) \in E \\ 0 & \text{otherwise} \end{cases}$$

- This matrix describes the probability of transitioning between nodes in a random walk on V. In particular, $P(i,j)$ is the probability of going from i to j in one time step.
- Note that although P is not symmetric, it is similar to the symmetric matrix $S = D^{-1/2}AD^{-1/2}$ via the transformation $S = D^{1/2}PD^{-1/2}$.

Definitions (cont.)

- Finally, define the Random Walk Laplacian of the graph G, as:

$$L = I - D^{-1}A = I - P$$

That is,

$$L(a,b) = \begin{cases} 1 & \text{if } a = b \\ -w_{a,b}/d(a) & \text{if } a \neq b \text{ and } (a,b) \in E \\ 0 & \text{otherwise} \end{cases}$$

Note that this definition is different from the standard Laplacian, which is defined as $L = D - A$

Some properties of L and P

- L, P are matrices of dimension $V \times V$.
- The sum of the elements on each row of P (respectively, L) is equal to 1 (respectively, 0).
- P has V real eigenvalues, counting multiplicity. (Proof: apply the Spectral theorem on S)
- P always has 1 as its eigenvalue, and any eigenvalue λ of P satisfies $|\lambda| \leq 1$. (Perron-Frobenius theorem)
- If (λ, v) is an eigenpair of P then $(1 - \lambda, v)$ is also an eigenpair of L, and vice versa.

Some properties of L and P (cont.)

- Consider the quantity $Lx = (I - P)x = x - Px$:

$$\begin{aligned} Lx(i) &= x(i) - \sum_{(i,j)\in E} P(i,j)x(j) \\ &= \sum_{(i,j)\in E} P(i,j)(x(i) - x(j)) \end{aligned}$$

- If $Lx(i) > 0$: (1) The value at node i is strictly greater than the weighted average of its neighbors. (2) Overall $x(i)$ loses value to neighbor nodes after a step of propagation.
- If $Lx(i) < 0$: (1) The value at node i is strictly less than the weighted average of its neighbors. (2) Overall $x(i)$ gains value from neighbor nodes after a step of propagation.

Some properties of L and P (cont.)

Key Takeaway

- The Random Walk Laplacian (L) acts as a discrete derivative of the graph, measuring the fluctuation of a node at each time step.
- The Walk Matrix (P) acts as an operator that carries out the propagation process.

Connection to the continuous Laplacian

Definition

Given a twice-differentiable, real-valued function f, the Laplace–Beltrami operator (or the Laplacian) of f is defined as:

$$\Delta f = \nabla^2 f$$

where

$$\nabla = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n} \right)$$

Alternatively, one can write the Laplace–Beltrami operator of f as:

$$\Delta f = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}$$

Connection to the continuous Laplacian (cont.)

For each axis x_i, the second derivative $\frac{\partial^2 f}{\partial x_i^2}$ acts as a local convexity test along that axis:

- $\frac{\partial^2 f}{\partial x_i^2} > 0$ (Convex): The value at this point is lower than the average of its immediate neighbors. [3]
- $\frac{\partial^2 f}{\partial x_i^2} < 0$ (Concave): The value at this point is higher than the average of its immediate neighbors.

Sounds familiar?

[3] A direct consequence of Jensen's inequality.

Connection to the continuous Laplacian (cont.)

Key takeaway

The Random Walk Laplacian is the discrete analogue of the continuous Laplace–Beltrami operator.

While the latter acts on functions defined over a continuous manifold, the former acts on functions defined over a graph constructed from discrete points sampled from that manifold.

Example: The Circle vs. The Cycle Graph

Continuous: A Circular String

- Domain: Angle $\theta \in [0, 2\pi)$
- Laplacian: $\Delta f = \frac{d^2 f}{d\theta^2}$
- Eigenfunctions: Smooth waves like $\sin(k\theta), \cos(k\theta)$

Discrete: A Cycle Graph (C_n)

- Domain: n points sampled on the circle
- Random Walk Laplacian (i-th row):
  $Lf(i) = \frac{1}{2}\left(2f(i) - f(i-1) - f(i+1)\right)$
- This is exactly the discrete approximation of $-\frac{d^2 f}{d\theta^2}$, up to a constant.

Diffusion Map

Motivation

- High-dimensional data often resides on a lower-dimensional, curved surface called a manifold. (Manifold Hypothesis)
- **Example:** The Swiss Roll dataset. It is a 2D sheet rolled up into 3D space.

Figure: The Swiss Roll dataset

Motivation (cont.)

- **The Problem:** Standard algorithms (like K-Means or PCA) rely on Euclidean distance, which does not reflect the true geometry of the dataset.

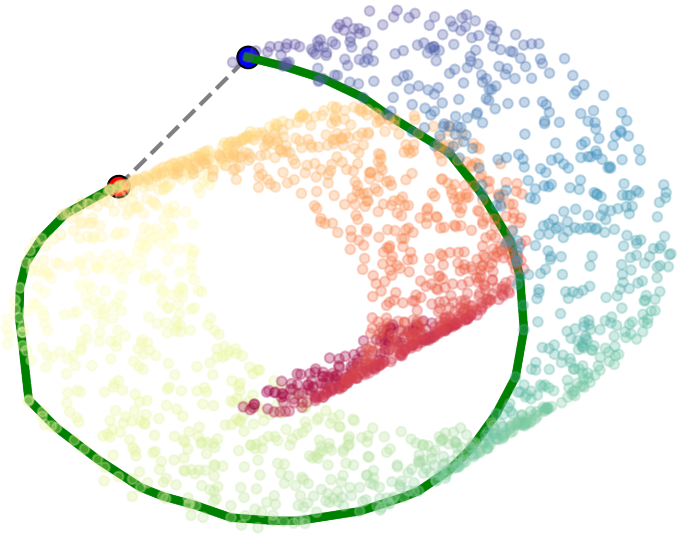

Figure: Euclidean distance (dotted) vs True distance (green)

Solution: Diffusion Maps

The Intuition

- In a real life object, heat, energy, etc... travels much more effectively from a region to nearby regions, instead of jumping through gaps (e.g. air).
- Instead of trying to draw complex curves, we treat the probability of going from point A to point B as "heat".
- The time it takes for "heat" to transfer from A to B on a manifold reflects the length of the "true" path (i.e. the path that goes along the surface of the manifold).

Solution: Diffusion Maps (cont.)

- We assume our dataset lies on a low-dimensional manifold $\mathcal{M}$ (Manifold Hypothesis).
- We construct the Adjacency matrix A, using an exponential decay, rotation-invariant kernel h.
- From A, we can construct D, and then P. We then diffuse P.

$\rightarrow$ The probability of diffusing from A to B gives us a robust measure of their true connectivity on $\mathcal{M}$.

Construction of the family of diffusions

- In real-world datasets, points are rarely sampled uniformly. Some regions will be densely packed, while others are sparse.
- If not accounted for, the random walks starting from inside the dense regions will not be able to propagate out.
- Suppose q is the sampling density function [4]. We introduce a parameter α to control the influence of q.

[4] Here we assume that the dataset is sampled across the entire manifold.

Construction of the family of diffusions (cont.)

Construction of the family of diffusions (Coifman and Lafon 2006)

(1) Fix $\alpha \in \mathbb{R}$ and a rotation-invariant kernel $k_\varepsilon(x,y) = h\left(\frac{\|x-y\|^2}{\varepsilon}\right)$.

(2) Let

$$q_\varepsilon(x) = \int_X k_\varepsilon(x,y) q(y)\,\mathrm{d}y$$

and form the new kernel

$$k_\varepsilon^{(\alpha)}(x,y) = \frac{k_\varepsilon(x,y)}{q_\varepsilon^\alpha(x) q_\varepsilon^\alpha(y)}.$$

(3) Apply the weighted graph Laplacian normalization to this kernel by setting

$$d_\varepsilon^{(\alpha)}(x) = \int_X k_\varepsilon^{(\alpha)}(x,y) q(y)\,\mathrm{d}y$$

and by defining the anisotropic transition kernel

$$p_{\varepsilon,\alpha}(x,y) = \frac{k_\varepsilon^{(\alpha)}(x,y)}{d_\varepsilon^{(\alpha)}(x)}.$$

Properties of the family of diffusions

- Define $E_K = \text{Span}\{\phi_l, 0 \leq l \leq K\}$ be the linear span of the first $K+1$ eigenfunctions of the Laplace–Beltrami operator Δ. These eigenfunctions satisfies the homogeneous Neumann condition: $\partial\phi_l/\partial\mathcal{M} = 0$, i.e. the boundary of $\mathcal{M}$ is an insulated wall, no probability can leak out.
- Furthermore, let:

$$L_{\varepsilon,\alpha} = \frac{I - P_{\varepsilon,\alpha}}{\varepsilon}$$

be the infinitesimal generator of the Markov chain. [5]

[5] Intuitively, one can think of $L_{\varepsilon,\alpha}$ as the derivative of $P_{\varepsilon,\alpha}$. Thus, $L_{\varepsilon,\alpha}$ dictates the behavior of $P_{\varepsilon,\alpha}$ at a given instance.

Properties of the family of diffusions (cont.)

Theorem (Coifman and Lafon 2006)

For a fixed $K > 0$, we have on E_K:

$$\lim_{\varepsilon \to 0} L_{\varepsilon,\alpha} f = \frac{\Delta(fq^{1-\alpha})}{q^{1-\alpha}} - \frac{\Delta(q^{1-\alpha})}{q^{1-\alpha}} f$$

Properties of the family of diffusions (cont.)

To recover the Riemannian geometry of the data set, we focus on the case $\alpha = 1$. Specifically:

$$\lim_{\varepsilon \to 0} L_{\varepsilon,1} = \Delta$$

In this case, the infinitesimal generator of the Markov chain approximates the Laplace–Beltrami operator on $\mathcal{M}$, regardless of q. Therefore, as $\varepsilon \to 0$, our diffusion process becomes the diffusion process on the surface of $\mathcal{M}$.

Properties of the family of diffusions (cont.)

As a consequence, our random walks will also approaches the Brownian motion on the surface of $\mathcal{M}$.

Thus, at a time step t, if two rows x and y of P^t share similar distributions (i.e. they jump around points on $\mathcal{M}$ in a similar manner), their distance on the surface of $\mathcal{M}$ must be small.

This is the main motivation for the **Diffusion Distance**.

Approximation on the finite dataset

In practice, it is impossible to obtain the formula for $q(y)$. Therefore, we can only approximate the kernel based on sampled data:

$$\bar{q}_\varepsilon(x_i) = \sum_{j=1}^{m} k_\varepsilon(x_i, x_j), \qquad \bar{d}_\varepsilon^{(\alpha)}(x_i) = \sum_{j=1}^{m} \frac{k_\varepsilon(x_i, x_j)}{\bar{q}_\varepsilon^{\alpha}(x_i)\, \bar{q}_\varepsilon^{\alpha}(x_j)}.$$

$$\bar{p}_{\varepsilon,\alpha}(x_i, x_j) = \frac{k_\varepsilon(x_i, x_j)}{\bar{d}_\varepsilon^{(\alpha)}(x_i)}, \qquad \bar{P}_{\varepsilon,\alpha} f(x_i) = \sum_{j=1}^{m} \bar{p}_{\varepsilon,\alpha}(x_i, x_j)\, f(x_j).$$

Approximation on the finite dataset (cont.)

This approximation method raises two problems:

- The dataset is finite, so there will always be a certain error in approximating P. Specifically, with high probability:

$$\left|\bar{P}_{\varepsilon,\alpha} f(x_i) - P_{\varepsilon,\alpha} f(x_i)\right| = \mathcal{O}\left(m^{-1/2}\varepsilon^{-d/4+1/2}\right)$$

  and:

$$\left|\bar{L}_{\varepsilon,\alpha} f(x_i) - L_{\varepsilon,\alpha} f(x_i)\right| = \mathcal{O}\left(m^{-1/2}\varepsilon^{-d/4-1/2}\right)$$

Criterion: In order to achieve a given precision with high probability, the number m of sample points must grow faster than $\varepsilon^{-d/4-1/2}$, where d is the dimension of the manifold.

Approximation on the finite dataset (cont.)

- The sampled points may not lie on the manifold, and are instead offset by a noise function η:

$$k_\varepsilon(x+\eta(x), y+\eta(y)) = k_\varepsilon(x,y) + \mathcal{O}\left(\frac{\|\eta\|}{\sqrt{\varepsilon}}\right)$$

and as a consequence, the perturbation of $P_{\varepsilon,\alpha}$ is the same order:

$$\tilde{P}_{\varepsilon,\alpha} = P_{\varepsilon,\alpha} + \mathcal{O}\left(\frac{\|\eta\|}{\sqrt{\varepsilon}}\right)$$

Criterion: The approximation is valid as long as the scale parameter $\sqrt{\varepsilon}$ remains larger than the size of the perturbation.

Application and Implementation

Spectral Embedding

Suppose we are working on dataset X. For the sake of simplicity, let's assume that:

1. X is finite with $|X| = m$, and
2. The constructed graph on X is connected.

Let π be the stationary distribution of P:

$$\pi(x) = \frac{d(x)}{\sum_{z \in X} d(z)}$$

Remind that if we set $S = D^{1/2} P D^{-1/2}$ then S is symmetrical, and therefore has m real eigenpairs (λ_i, ϕ_i) $(0 \leq i \leq m-1)$ where $\{\phi_i\}$ is an orthonormal basis.

Note: We are working on P, but this argument also works for $P_{\varepsilon,\alpha}$, and its discrete counterpart $\bar{P}_{\varepsilon,\alpha}$.

Spectral Embedding (cont.)

Therefore at a time t one can write:

$$S_t = \sum_{i=0}^{m-1} \lambda_i^t \phi_i \phi_i^\top$$

and specifically:

$$S_t(x, y) = \sum_{i=0}^{m-1} \lambda_i^t \phi_i(x) \phi_i(y)$$

Spectral Embedding (cont.)

From here, we can recover P:

$$P_t(x,y) = \sum_{i=0}^{m-1} \lambda_i^t \psi_i(x)\varphi_i(y)$$

where $\psi_i(x) = \phi_i(x)/\sqrt{\pi(x)}$ and $\varphi_i(y) = \sqrt{\pi(y)}\phi_i(y)$. For a fixed x, this can be rewritten as:

$$\begin{bmatrix} \varphi_0 & \varphi_1 & \dots \varphi_{m-1} \end{bmatrix} \begin{bmatrix} \lambda_0^t \psi_0(x) \\ \lambda_1^t \psi_1(x) \\ \dots \\ \lambda_{m-1}^t \psi_{m-1}(x) \end{bmatrix} = \begin{bmatrix} P_t(x,0) \\ P_t(x,1) \\ \dots \\ P_t(x,m-1) \end{bmatrix} = P_t(x,\cdot)$$

Spectral Embedding (cont.)

By exploiting the orthonormality of the ϕ_i basis, we can define the Diffusion Distance between two distributions $P_t(x,\cdot)$ and $P_t(z,\cdot)$ as follow:

$$\begin{aligned} D_t(x,z)^2 &= \|P_t(x,\cdot) - P_t(z,\cdot)\|^2_{1/\pi} \\ &= \sum_{y\in X} \frac{(P_t(x,y) - P_t(z,y))^2}{\pi(y)} \\ &= \sum_{y\in X} \left(\sum_{i=0}^{m-1} \lambda_i^t (\psi_i(x) - \psi_i(z))\phi_i(y) \right)^2 \\ &= \sum_{i=0}^{m-1} \lambda_i^{2t} (\psi_i(x) - \psi_i(z))^2 \end{aligned}$$

Note: π is just d, scaled by a constant factor. Thus, in real implementations, we simply use d instead of π.

Choosing the right ε

(Coifman, Shkolnisky, et al. 2008) have shown that, for small values of ε, the sum of the weight matrix approximates the manifold's volume:

$$\log\left(\sum_{i,j} W_{i,j}(\varepsilon)\right) \approx \frac{d}{2}\log\varepsilon + \log\left(\frac{N^2(2\pi)^{d/2}}{\mathrm{vol}(\mathcal{M})}\right)$$

By calculating the empirical sum across a range of ε values, we take the derivative with respect to $\log\varepsilon$. The unique maximizer ε_0 of this derivative identifies the optimal scale where the discrete data best approximates the continuous manifold.

The Berry-Giannakis-Harlim (BGH) Algorithm

Objective: Find the optimal bandwidth ε_0 and intrinsic dimension d.

1. **Define a Search Grid:** Create a logarithmically spaced array of candidate values for ε (e.g., 2^{-10} to 2^{10}).
2. **Calculate the Matrix Sums:** For each candidate ε_k, construct the weight matrix $W(\varepsilon_k)$ and compute its grand sum:

$$S(\varepsilon_k) = \sum_{i,j} \exp\left(\frac{-\|x_i - x_j\|^2}{2\varepsilon_k}\right)$$

3. **Compute the Log-Derivative:** Calculate the discrete derivative (slope) of the log-sum with respect to $\log \varepsilon$:

$$L(\varepsilon_k) = \frac{\Delta \log S(\varepsilon_k)}{\Delta \log \varepsilon_k}$$

4. **Extract the Peak:** Find the ε_0 that maximizes $L(\varepsilon)$.
5. **Estimate Dimension:** The peak slope directly yields the intrinsic dimension: $d \approx 2 \times \max L(\varepsilon)$.

Implementation

For detailed implementation, see:

https://github.com/BanAnA9205/Diffusion-Map

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